

A Composable Language for Governed Agent Work

This is the high-level system view. It explains the shape and stakes of the proposed system for readers who do not yet know its vocabulary. Terms remain provisional until an ontology view owns them, and load-bearing choices remain open until an engineer view owns their verdicts.

1. About this document

This document presents a high-level view of a possible language for coordinating work across people, agents, tools, and records. It examines how a human request can remain understandable as it is interpreted, revised, delegated, acted on, and reviewed—and how its context, decisions, evidence, authority, and history can remain connected throughout that process.

This is a proposal. Its concepts and definitions are not final and should be revised when better alternatives emerge.

The document begins with how requests are made and followed. It then examines how work can be described, connected, delegated, and observed. Finally, it considers the supporting architecture, the decisions that remain open, and which parts might benefit from mathematical formalization.

Alternative framings considered

Framing	Why it is insufficient on its own
A better task manager	It does not explain evolving definitions, authority, evidence, or recursive composition.
A multi-agent workflow engine	It starts too late: the meaning and legitimacy of the work are already assumed.
An ontology platform	It may describe the world without governing effects, execution, and live work.
A document methodology	It cannot by itself make runtime state, capabilities, and provenance observable.

2. What must remain visible

The system should make it possible to understand a piece of work without reconstructing it from folder names, disconnected documents, agent prompts, and memory. A reader should be able to examine its origin, current state, relationships, authority, and history from the records that describe it.

The system should connect abstract statements about work to the records, evidence, decisions, observations, procedures, or enforcement mechanisms that make them inspectable. Concrete does not simply mean more detailed or implemented in code; it means that the basis for a statement can be found and examined. Understanding a piece of work therefore requires four connected kinds of information.

Origin and meaning

- What was requested?
- How has that request been interpreted, and which interpretation is current?
- Which assumptions, definitions, and constraints shape that understanding?
- Which earlier or competing interpretations remain relevant, and why did the current interpretation prevail?

State and authority

- What is only being discussed or proposed?
- What has been accepted?
- What has been authorized for execution?
- What is waiting, blocked, declined, cancelled, or complete?

These distinctions matter because a detailed proposal is not automatically an accepted decision, and an accepted decision is not automatically permission to act.

Delegation and activity

- Which person or agent received the work?
- What objective, context, tools, limits, and authority did they receive?
- Which larger request, investigation, plan, specification, or decision does the activity support?
- What has started, and what never launched?

These questions make concrete activity traceable to the purpose and authority behind it.

Evidence and change

- Which evidence supports a claim or decision?
- What result was produced?
- Which description, decision, artifact, or state changed because of that result?
- Has the result been verified or accepted as establishing completion, does it challenge an assumption, or does it create another claim to review?

The system should also make missing or partial grounding visible. A plan may lack authority. A claim may have incomplete evidence. A requirement may lack implementation. A metric may exist without an instrument, or may depend on human observation rather than automated measurement. These are valid states when they are explicit; they become misleading when the system fills the gaps through inference.

Any later view can display only the distinctions that the underlying system preserves. If proposals and authorizations share one state, if names are treated as identity, or if results are stored without their origin, no interface can faithfully recover what was never recorded.

The next section examines the basic distinctions needed to preserve this information.

Alternative framings considered

Framing	Why set aside
Add a better status dashboard	A view cannot recover distinctions that the underlying records never preserved.
Store every event in one log	Sequence alone does not explain meaning, authority, evidence, or semantic relationships.
Ask agents to summarize the current state	A generated summary is another assertion; it does not replace attributable source records.

3. Objects, descriptions, and hypotheses

The working model separates identity, description, relationship, and operational acceptance. Objects remain addressable while their names, descriptions, classifications, locations, or relationships change. Tags allow low-cost descriptions before the system or its users know enough to commit to a stronger type or rule.

Descriptions are hypotheses rather than timeless truth. A person or agent may assert that a construct is valid, useful, related, complete, or ready. The assertion remains distinguishable from its evidence, its contextual acceptance, and any authority to produce effects.

This distinction makes progressive definition possible:

```
partially described object
-> additional tags and observations
-> candidate properties and relations
-> reviewed or accepted contracts
-> operational use under explicit authority
```

The arrows do not mean that every object must follow one maturity ladder. They show increasing commitment that a workflow profile may govern.

Alternative framings considered

Framing	Why set aside
Require complete schemas at creation	It makes early capture expensive and encourages users to keep knowledge outside the system.
Treat all tags as authoritative facts	Cheap description would silently acquire operational force.
Make names or paths the identity	Renaming, moving, and presenting alternative views would rewrite meaning.

4. Relations and composition

Relationships are first-class because several different meanings can connect the same two objects. A Dispatch may be generated from another object, authorized by a confirmation, contained in a Plan, and reviewed by a separate process. None of those relationships safely implies the others.

Accepted relations need enough semantics to support validation and derived views. The research program currently expects relation contracts to address type, direction, source, destination, scope, version, provenance, cardinality, transitivity, inheritance, and cycle policy. Different relation types may form trees, forests, DAGs, or controlled cyclic graphs.

Relations can also compose. A useful composition must preserve the path that justified a derived claim or action. The system should be able to explain whether a result follows from direct relations, a permitted transitive closure, a translation between vocabularies, or a user-approved rule.

Physical placement is one projection of these properties and relationships. The same objects may appear in a repository-root view, an `internal_tools` view, a project view, a research view, or a task-oriented dashboard without acquiring different identities merely because they appear in different places.

5. A recursive grammar of work

The familiar sequence below is a useful first reading:

```
Intent
-> Plan
-> Research
-> Discovery / Design
-> Spec
-> Code
-> Verification
```

It is not a universal irreversible pipeline. It is a candidate grammar of work kinds whose instances can recur at different scopes and connect through typed relations.

```
flowchart LR
  I[Intent] --> P[Plan]
  P --> R[Research]
  R --> D[Discovery / Design]
  D --> S[Spec]
  S --> C[Code]
  C --> V[Verification]

  R -. investigates or challenges .-> P
  P -. requests .-> R
  S -. exposes a local planning need .-> P2[Local Plan]
  V -. reopens .-> R2[Research]
  V -. reopens .-> S
```

Research for a Plan is therefore ordinary, not exceptional. A Plan can identify uncertainty and request Research. Research can evaluate the assumptions, feasibility, precedents, or consequences of that Plan. Its findings may support the Plan, revise it, split it, or make it unnecessary.

A workflow profile may still require a default progression. It can state which work kinds, evidence, reviews, and approvals are required before a more expensive or authoritative commitment. The invariant candidate is not one global order. It is that the applicable profile, transitions, evidence, authority, bypasses, and

reopenings are explicit and historically reconstructable.

Alternative framings considered

Framing	Why set aside
One global lifecycle	It cannot naturally represent Research for a Plan or local planning inside a Spec.
Completely free graph	It makes obligations, promotion, and completion too easy to evade.
Folder nesting as workflow	It confuses navigation with semantic and operational relationships.

6. Recursive work without recursive orchestrator authority

Recursive structure does not imply recursive orchestration. An invoked orchestrator must not invoke another orchestrator. Instead, the root orchestrator resolves the work graph and materializes small WorkPackages for leaf agents.

Each assignment should receive only what its task needs: an objective, selected context, source responsibilities, a small tool profile, a budget, an expected output, applicable gates, and explicit authority boundaries. A parent relationship does not automatically transmit tools, budget, evidence, approval, or terminal state.

This allows deep work structures with shallow execution authority:

```
recursive work graph
-> root orchestration and confirmation
-> bounded leaf assignments
-> attributable outputs and observations
-> reconstructed progress over the original graph
```

7. Plans as revisable priors

The canonical Plan contract owns the term and its authority boundary. At this narrative altitude, a Plan records the best current route, not an immutable prediction of the future. Research may change its assumptions. New constraints may move its stopping point. Implementation may expose a missing Design decision. Verification may show that the Spec or the Plan needs to be reopened.

Revisions should create new attributable facts and versions rather than silently rewriting what was previously believed. The system can then distinguish failure to follow a Plan from justified revision of the Plan.

Every Plan searches for governing authority, but that search may end without finding one. An unresolved Plan remains a visible, revisable proposal; it cannot become a binding route or execution authority. Authorship, parentage, physical placement, and apparent completeness do not fill the authority gap.

Program phases work similarly. A phase groups questions, dependencies, and readiness conditions that are useful to consider together. It is not necessarily a permanent product layer. Phases may overlap, split, merge, move, or be superseded as the program learns.

8. From a request to observable work

A possible operational shape is:

```
conversation statement
-> candidate WorkIntent
-> admitted WorkIntent
-> Plan / Research / WorkPackage / DispatchCandidate
-> confirmed Dispatch
-> TaskRun / AgentAttempt
-> append-only lifecycle facts
-> current-work, audit, and reminder projections
```

The original request is not mutated into a status field. Accepted facts record that something was admitted, confirmed, started, blocked, completed, cancelled, or never launched. Read models derive the current view and expose freshness when the system does not know what happened recently.

This separation matters because a dashboard must not create authority by displaying a button, status, or inferred relationship. It presents accepted facts and identified uncertainty.

9. The kernel is a research question

“Kernel” currently means a possible minimal contract that allows independently evolving parts of the system to interoperate and remain auditable. It does not yet name a selected component.

The first research phase must compare:

- one universal kernel;
- several specialized kernels connected by protocols;
- a microkernel with extensions;
- a kernel-of-kernels that governs compatibility and translation;
- or an architecture in which “kernel” is not a useful organizing concept.

Candidate invariants include identity, versioning, provenance, relative validity, typed relations, explicit authority, traceable composition, and historical preservation. Each remains a hypothesis until research tests its necessity, scope, compatibility with the others, and behavior under composition.

The phrase **kernel-of-kernels** currently covers four questions that must remain separable: what makes a kernel declaration governable, which laws are genuinely global, how independently valid kernels compose, and which small checker evaluates those claims. One mechanism might eventually implement several responsibilities, but the architecture does not yet assume that they are one concept.

Lean is included because it is already important in the surrounding work. Its actual theorem-prover kernel is also a useful **analogy-only** precedent: rich syntax, tactics, macros, and elaboration may remain extensible because their output is checked by a much smaller trusted type checker. This suggests a product shape in which flexible authoring sits outside a compact acceptance boundary that checks explicit evidence.

The analogy has a hard boundary. Lean's kernel checks elaborated declarations and proof terms against its core type theory and accepted environment. It does not decide product authority, provenance, operational precedence, temporal obligations, or external effects, and even well-typed axioms may be mutually

inconsistent. The project must therefore separate Lean as explanatory notation, Lean as a checker or generator of selected artifacts, Lean-checked results as governed evidence, and the runtime mechanisms that actually enforce authority. The `permguard` artifact elsewhere in the repository is a policy decision program verified in Lean and informally called a “Lean kernel”; it is not the Lean prover’s own trusted kernel.

Alternative framings considered

Framing	Why it is insufficient on its own
Use the Lean kernel as the product metakernel	It checks core type-theoretic declarations, not product authority, provenance, temporal rules, conflict resolution, or effects.
Call the invariant metadata schema the metakernel	A complete record can still describe a false, contradictory, unenforced, or unauthorized invariant.
Resolve every conflict by authority or priority	Authority may be out of scope or incomparable, and priority cannot make incompatible global laws jointly satisfiable.

10. Candidate architecture by responsibility

The architecture can be understood as collaborating responsibilities rather than fixed product layers:

1. **Authoring and interaction** captures user intent, incomplete descriptions, questions, proposals, and decisions.
2. **Knowledge and relationship management** preserves identities, properties, tags, typed relations, definitions, provenance, and alternative projections.
3. **Governance and authority** distinguishes claims, evidence, acceptance, promotion, confirmation, capabilities, and effect authorization.
4. **Planning and orchestration** resolves recursive work graphs into bounded, confirmable assignments and schedules.
5. **Execution** invokes tools and agents under materialized capabilities and records attempts.
6. **Event history and projections** preserve accepted lifecycle facts and build current views.
7. **Verification and formalization** check selected contracts through runtime validators, tests, reviews, and potentially Lean.
8. **Observability and product views** expose pending, active, blocked, completed, abandoned, and unknown work at multiple levels.

These responsibilities may be deployed together or separately. The research must determine which boundaries are semantic, which are authority boundaries, and which are merely implementation choices.

Alternative framings considered

Framing	Why set aside
Declare these as permanent layers	The layer model itself remains an open, configurable projection.
Put all semantics in one central service	It may turn a minimal interoperability contract into a product monolith.
Let every subsystem define everything locally	Translation, audit, and authority could become implicit or contradictory.

11. Candidate tool families

Tool selection comes after responsibility boundaries. The following is an option inventory, not a recommendation:

Responsibility	Candidate options to compare	Main research question
Identity, metadata, and transactional authority	PostgreSQL; document databases; append-oriented stores	Which facts require transactions, constraints, and authoritative ownership?
Relationship traversal and alternative views	Relational edge tables; graph databases such as Neo4j; derived graph indexes	Is graph traversal authoritative or a rebuildable projection?
Durable event history	Database outbox/event tables; dedicated event stores; Kafka- or NATS-style streams	Which events are authoritative, and which transport observations are replayable?
Workflow orchestration	Purpose-built runtime; Temporal-style workflow engines; Dagster/Prefect-style orchestrators	Can the engine preserve confirmation, capabilities, recursive work, and explicit authority?
Policy and authorization	Application-owned policies; OPA; Cedar-style policy evaluation	How are policy versions and decisions bound to an execution attempt?
Search and retrieval	Database search; Elasticsearch/OpenSearch-style indexes; embedding-assisted retrieval	Which retrieval results are evidence, and how is freshness shown?
Observability	OpenTelemetry-compatible traces and metrics; journal-derived operational views	How are rich histories separated from bounded-cardinality health signals?
Formalization and validation	Lean; schema validators; property tests; model checking	Which claims deserve formalization, and how do formal artifacts connect to runtime evidence?
Artifact history and collaboration	Git; database versions; hybrid GitOps	Which changes need branching and review, and which live facts cannot wait for commits?

The final architecture may use several tools for one responsibility or one tool for several responsibilities. The selection criterion is whether authority, provenance, replay, and interoperability remain explicit—not whether a tool can store a similar shape.

12. A representative walkthrough

Suppose a user asks: “Plan how to introduce a new governed research capability.”

1. The statement is captured as a candidate WorkIntent.
2. A Plan is created as a revisable prior.
3. The Plan identifies uncertainty and requests Research about existing capabilities.
4. The Research targets the Plan, returning evidence that revises its assumptions.
5. Discovery/Design synthesizes the evidence into alternatives and explicit open decisions.
6. A Spec records an accepted contract for one bounded capability.
7. The root orchestrator prepares a DispatchCandidate for implementation and review.
8. If it waits for confirmation, it remains visible and can trigger a reminder.

9. Once confirmed, the runtime creates bounded agent attempts with explicit context and tools.
10. Code is linked to the Spec version it attempts to realize.
11. Verification checks the implementation and may reopen Research or Spec.
12. The dashboard reconstructs current state from accepted lifecycle facts while preserving the complete history of revisions and non-launched candidates.

The walkthrough is intentionally recursive. Research did not merely follow the Plan; it acted on the Plan and changed it.

13. What remains open

The research program must still determine:

- whether stable object identity survives changes to function or objective;
- whether Plan is the only route-bearing object allowed to remain durable while governing authority is absent, unknown, or contested;
- which work names are artifact kinds, activity kinds, session kinds, or contextual roles;
- the typed relation grammar and permitted recursive compositions;
- how workflow profiles require, skip, or reopen work;
- whether `Spec` and `Code` are the only universally required work kinds, with the route through Plan, Research, Experiment, Discovery, Design, Verification, or other kinds selected through a user-confirmed installation profile;
- whether installation is the right confirmation boundary for that profile or whether the route must remain adjustable per project, risk, and context;
- whether there is one kernel, several, or none;
- whether a meta-contract, global invariant set, composition protocol, and trusted bootstrap need separate owners or one bounded mechanism;
- the boundary between loose tags and accepted operational relations;
- how candidate validity becomes contextual acceptance or executable authority;
- how pending Dispatch candidates are registered and reminded without entering the executable ledger;
- which semantics Lean should own, check, generate, or merely explain;
- and which infrastructure choices preserve these distinctions with acceptable complexity.

Every governed explanatory, research, planning, design, specification, formalization, review, or decision artifact must retain an explicit `Open Questions` section. An empty section records that no questions are currently known; it does not assert completeness. Resolved or deferred questions remain in history with their later status.

14. Named architectural stances

This view names but does not decide the following tensions:

- `stance:meta-invariant-boundary` → `engineer-view#decision-meta-invariant-boundary`
- `stance:trusted-bootstrap-boundary` → `engineer-view#decision-trusted-bootstrap-boundary`

- `stance:kernel-topology` → `engineer-view#decision-kernel-topology`
- `stance:plan-authority-exception-scope` → `engineer-view#decision-plan-authority-exception-scope`
- `stance:recursive-work-grammar` → `engineer-view#decision-recursive-work-grammar`
- `stance:workflow-profile-authority` → `engineer-view#decision-workflow-profile-authority`
- `stance:minimal-universal-work-contract` → `engineer-view#decision-minimal-universal-work-contract`
- `stance:dispatch-candidate-registry` → `engineer-view#decision-dispatch-candidate-registry`
- `stance:relation-authority` → `engineer-view#decision-relation-authority`
- `stance:projection-vs-authority` → `engineer-view#decision-projection-authority`
- `stance:lean-runtime-boundary` → `engineer-view#decision-lean-runtime-boundary`
- `stance:formalization-correspondence` → `engineer-view#decision-formalization-correspondence`
- `stance:storage-and-orchestration-tools` → `engineer-view#decision-tool-boundaries`

`Plan` defers to the canonical `Plan` contract. Other provisional terms used here defer to a future ontology view, including `WorkIntent`, `Research`, `Discovery`, `Design`, `Spec`, `WorkPackage`, `DispatchCandidate`, `Dispatch`, `TaskRun`, `WorkflowProfile`, `kernel`, `metakernel`, `conformance checker`, `global invariant`, `composition witness`, `bootstrap`, `relation`, `validity`, and `authority`.

15. What this view does not cover

This document does not own canonical definitions, record schemas, field-level contracts, failure codes, validator behavior, infrastructure selection, or implementation verdicts. Those belong in the future ontology and engineer views and in the governed research artifacts that precede them.

16. Mathematical and Lean formalization appendix

This is the first formal proposal generated by R1 of the Agent Work Language Research. Its research record preserves the independent returns; its findings preserve their synthesis. The formalization is a hypothesis and remains inside the existing system view rather than becoming an independent plan.

16.1 Status and correspondence

The appendix distinguishes definitions, premises, candidate axioms, invariants, propositions, countermodels, proof obligations, `proof-present-in-bound-source`, and `machine-checked-currently`. The last status is reserved for a later build and dependency audit. Mappings are classified as `direct`, `adapted`, `analogy-only`, `conflicting`, `insufficient`, or `no-correspondence`.

Three judgments must not be conflated:

$$\begin{array}{ll} \text{model soundness} & \Rightarrow \text{product correspondence} \\ \text{product correspondence} & \Rightarrow \text{execution authority} \end{array}$$

This reflects the responsibility boundaries in the ACI architecture: semantic structure, accepted runtime facts, derived projections, executable authority, and external effects are not the same thing.

16.2 Many-sorted carriers before a universal object

Let K be a provisional set of construct kinds. For each $k \in K$, let X_k be a carrier of constructs, I_k its identity space, D_k its description space, and Ver a version order. Define

$$\begin{array}{ll} \text{id}_k & : X_k \rightarrow I_k \\ \text{desc}_k & : X_k \times Ver \rightarrow D_k \end{array}$$

Identity is stable while versioned names, descriptions, tags, classification, lifecycle posture, and physical location may change. This corresponds to Section 3 and to stable `dispatch_id`, `run_id`, and artifact identities in the ACI domain.

Several carriers are used to avoid prematurely collapsing Entity, Assertion, Event, Definition, Relation, and Execution into one universal `Object`, a risk recorded in the foundational brief. This is an adapted model, not a selected product ontology. Tags remain low-cost incomplete assertions; attachment alone does not promote them to accepted typed properties.

16.3 Direct relations and witnessed composition

For each relation signature ρ , let

$$D_\rho(x, y)$$

be the type of accepted direct facts from x to y . A signature owns endpoint kinds, direction, scope, version, provenance requirements, cardinality, transitivity, inheritance, and cycle policy. These responsibilities originate in Section 4 and the research subplan.

A direct fact is different from a derived path:

$$\text{Path}(x, z, w, r)$$

where $w = (e_1, \dots, e_n)$ lists accepted direct facts and r identifies the rule and version that admitted the derivation. An explicit witness

$$\text{Adm}(\rho, \sigma ; \tau)$$

is required before relation kinds ρ and σ can compose into kind τ . Trees, DAGs, and controlled cycles therefore arise from relation-specific policies, not one universal hierarchy.

The base is a proof-relevant typed multigraph, not automatically a category. A relation family earns categorical structure only after identities, closure, composition, and coherence have been specified. The DomainSpec relationship model is useful typed-graph precedent, but its heterogeneous relation kinds do not already share a universal composition law.

Candidate invariant R1. Every derived relation retains its direct witnesses and the exact composition rule/version.

Non-claim. Reachability is not a direct fact, proof of truth, or delegation of authority.

16.4 History, provenance, and projections

Let `Prov(z)` be a typed origin and transformation chain. Provenance explains where `z` came from; it is not correctness, trust, or authority.

For aggregate `a`, let `H_a = [e_1, ..., e_n]` be its ordered accepted history and

```
fold_v : H_a -> S_a
```

be a pure reducer at version `v`. This maps directly to `RuntimeEventEnvelope` and replay in the ACI workflows.

Candidate invariant H1. The same ordered history, reducer version, and payload equality produce the same state without external effects. Corrections append rather than rewrite history.

For view kind `v`, define a rebuildable projection `pi_v : H -> V_v`. Folder trees, indexes, dashboards, and current-work views are projections. Physical location is therefore one property, not identity or semantic authority.

Candidate invariant V1. A projection may expose accepted authority evidence but cannot create authority. A stale dashboard showing `ready` while no accepted confirmation exists is the collapse test.

16.5 Authority, context, and effects

Authority is not ordinary graph reachability. Define

```
Auth(p, o, x, s, v, b)
```

to mean that principal `p` may perform operation `o` on `x`, in scope `s`, at version `v`, on accepted basis `b`. This is adapted from the explicit authority model in CAV2 and the confirmed-Dispatch boundary in the ACI domain.

Let `Materialize(d,a,m,c)` record that Dispatch `d`, for attempt `a`, received context `c` through explicit manifest `m`.

Candidate invariant A1. Abbreviate `HasAuth(y) := exists p,o,s,v,b. Auth(p, o, y, s, v, b)`, and let `HasTools`, `HasBudget`, and `HasEvidence` abbreviate the corresponding accepted records for `y`. Then

$$\text{Lineage}(x, y) \Rightarrow \text{HasAuth}(y) \text{ or } \text{HasTools}(y) \text{ or } \text{HasBudget}(y) \text{ or } \text{HasEvidence}(y)$$

The disjunction is load-bearing: lineage transmits none of the four individually, which is strictly stronger than merely failing to transmit all four at once.

This “no automatic inheritance” law concerns execution lineage. It is not a universal rule for documents: a typed document relation may define content inheritance, overlay, or precedence under its own policy. The parent may explicitly materialize selected context for a child, producing a versioned and attributable record.

External effects cross an explicit $\text{Fence}(d, r, m, e)$, corresponding to $\text{ExecutionAuthorityFence}$. Mathematics can describe required evidence, but cannot prove that a physical sandbox, process tree, credential boundary, or sole writer was enforced.

16.6 Recursive work with shallow orchestrator authority

Let

$$W = (V, E, \text{rank}, B)$$

be a finite typed work graph with rank or bounded rounds and budget B . Its nodes may be Intent, Plan, Research, Discovery/Design, Spec, Code, or Verification. This is not a linear pipeline: Research may challenge a Plan, a Spec may open local Research, and Verification may reopen Design.

Let $\text{compile}(D, W) = L$ map a confirmed root Dispatch and valid work graph to bounded leaf assignments.

Candidate invariant W1. Every leaf has explicit tools, context, budget, and gates, and no invoked orchestrator has an outgoing orchestrator-invocation edge.

This formalizes the user-confirmed constraint in Section 6 and the invocation brief. It does not forbid recursive documents, plans, feedback, or research. It separates semantic nesting from runtime authority nesting and does not claim termination of arbitrary agent behavior.

16.7 Bounded contracts and the kernel-of-kernels question

Let Sig_i be a versioned bounded contract and T_{ij} a declared translation. Define

$$\text{Compat}(\text{Sig}_i, \text{Sig}_j)$$

as a record of translations, preserved invariants, owner acceptance, conflicts, and residue. Calling T_{ij} a functor is justified only when both sides have the required category structure and identity/composition preservation is proved.

A finite bootstrap boundary is proposed as

$$B_0 = (\text{roots}, \text{versions}, \text{owners}, \text{validators}, \text{assumptions})$$

The current refinement separates five objects that the informal phrase `kernel-of-kernels` can otherwise collapse:

```
M    = meta-contract
G    = global invariant set
K_i  = bounded domain kernel
Q    = conformance checker
C_ij = composition witness
```

`M` states what evidence makes a kernel or invariant declaration well-formed: identity, version, owned scope, applicability, authority basis, evaluation semantics, dependencies, violation posture, provenance, and lifecycle are candidate responsibilities, not a settled record schema. `G` contains only the laws every accepted composition must preserve. Each `K_i` owns bounded domain semantics. `C_ij` records the translations, preserved laws, conflicts, assumptions, and residue for one declared composition. `Q` implements a small set of versioned judgments over these declarations and witnesses; it is not identical to the language `M` that they inhabit.

Writing `Ctx` for the accepted declaration context against which a judgment is evaluated, the intended judgment is closer to

```
B_0 ; Ctx  |-_Q  WellFormed_M(K_i)
B_0 ; Ctx  |-_Q  C_ij : Compatible_G(K_i, K_j)
```

than to `M |- K_i`. The meta-contract need not derive a domain kernel's rules or prove them jointly satisfiable. A compatibility witness is scoped evidence, not permission to execute. A separate effect boundary still requires accepted authority.

Authority, precedence, and logical compatibility are also different relations. Authority governs who may propose, accept, revise, or enforce a rule. Precedence selects among rules only where an accepted composition policy defines such selection. Incompatible invariants may require rejection, isolation, translation, version change, or escalation; a generic "higher authority wins" rule would allow local policy to weaken global law and is therefore a collapse test, not a candidate default.

The actual Lean kernel motivates only the shape of a small trusted checker beneath a richer elaboration layer. It does not supply these product semantics. Lean's kernel checks core terms relative to an environment and admitted axioms; this system still needs separate correspondence, governance, journal, and physical-enforcement boundaries. Independent rechecking is a useful precedent for reducing implementation trust, but not proof that the proposed `B_0`, `M`, or `C_ij` is sufficient.

Finite dependencies, well-founded rank, decidable checks, and explicit roots can make selected well-formedness and compatibility checks terminate relative to `B_0`. They do not self-justify those roots. Whether the product needs one kernel, several kernels, a microkernel, a metakernel, a composition protocol, or some combination remains a Phase 1 research question.

16.8 Proposition and countermodel ledger

ID	Claim	Responsibility	Status	Collapse test or boundary
P-01	derived paths retain direct witnesses and rule versions	relations	open	derived fact without reconstructible path
P-02	projections cannot manufacture authority	read models	open	projection-only state launches work
P-03	fixed replay is deterministic and effect-free	journal/reducer	open	reducer reads clock/provider or emits effects
P-04	confirmed Dispatch semantics are immutable under one identity	invocation	open	topology changes under same confirmed digest
P-05	lineage does not transmit authority or context	invocation/documents	open	child launches from ancestry alone
P-06	finite recursive work compiles to shallow authority	orchestration	open	an invoked orchestrator launches another orchestrator
P-07	policy concatenation is decision meet under the inspected algebra	permissions	proof-present-in-bound-source	matcher safety unproved; empty policy is <code>allow</code>
P-08	node-local data cannot characterize graph acyclicity	graph validation	proof-present-in-bound-source	unary checks all pass on a cycle
P-09	faithful schema translation need not preserve all instances	translation	proof-present-in-bound-source countermodel	analogy-only until product mapping exists
P-10	finite bootstrap checking terminates relative to roots	compatibility	open	unranked dependency cycle
P-11	meta-well-formedness does not imply invariant preservation or execution authority	kernel governance	open	complete metadata accepted as proof or permission
P-12	accepted kernel composition preserves global invariants under its declared witness	composition	open	individually valid kernels compose into a globally invalid state
P-13	authority and precedence are scope-qualified partial relations	governance	open	globally stronger rule inferred from unrelated or out-of-scope authority
P-14	checker acceptance and effect enforcement remain separately evidenced	runtime correspondence	open	accepted certificate while the physical enforcement point is absent or bypassed
P-15	independent checker implementations agree on the accepted core	bootstrap trust	open	same normalized obligation receives divergent verdicts

The bounded Lean observations for P-07 through P-09 are recorded in the research evidence. No build was run.

16.9 Lean roadmap

The Lean model should encode only the accepted pure subset:

1. claim status and `CorrespondenceRecord`;
2. `Kind`, `ObjId`, `Version`, and versioned descriptions;
3. `RelationSig`, `DirectEdge`, `ComposeWitness`, `DerivationPath`, and `CyclePolicy`;
4. `AcceptedEvent`, `History`, pure `fold`, and rebuildable `Projection`;
5. `AuthorityClaim`, `ContextManifest`, reveal evidence, and abstract fence evidence;
6. finite `WorkGraph`, rank/budget, compiler validity, and `LeafAssignment`;
7. bootstrap, meta-contract, conformance judgments, bounded domain contracts, global-law preservation, translations, composition witnesses, and residue; and
8. counterexamples before broader categorical enrichment.

The first dependency cone should prefer structures, indexed families, relations, lists, finite graphs, and predicates. Kan extensions, Yoneda, fibrations, sheaves, operads, thermodynamic metaphors, and reflection towers remain outside the core unless a smaller structure fails a named obligation and an infrastructure owner establishes correspondence.

Before any result becomes `machine-checked-currently`, a verification record must name the Lean project/toolchain, build target, dependency cone, successful build, `sorry` and axiom audit, source digest, correspondence review, and relationship to runtime validators. Proof remains evidence, never execution authority.

16.10 Residue and non-claims

- No universal `Object` or category is selected.
- No composition law applies to every relation.
- No metakernel, global invariant set, or composition protocol is selected.
- No well-formed invariant record proves that its predicate is preserved.
- No authority or precedence relation repairs a logical contradiction by itself.
- No proof establishes that deployed implementation matches the model.
- No projection establishes freshness, completeness, or causal truth.
- No provenance or schema-validity record establishes correctness, promotion, or authority.
- No lineage edge delegates tools, evidence, budget, approval, or terminal state.
- No theorem models physical enforcement, provider behavior, credentials, or writer safety.
- No termination theorem covers agents, reviews, reminders, retries, or feedback loops.
- "Plan as revisable prior" remains a metaphor until an update calculus is selected.
- Layers and phases remain configurable views, not universal invariants.

16.11 Open Questions

ID	Question	Status	History
ALF-OQ-001	What are the minimal primitive carriers, and can a common <code>Object</code> avoid collapse?	open	Many-sorted start proposed 2026-07-24.
ALF-OQ-002	Which relation kinds admit composition, partial composition, or none?	open	Universal composition rejected 2026-07-24.
ALF-OQ-003	Which derived closures may affect decisions, under what witness/version contract?	open	Opened 2026-07-24.
ALF-OQ-004	Which lifecycle model combines accepted history, transitions, intervals, and temporal obligations?	open	Opened 2026-07-24.
ALF-OQ-005	What is the minimal authority record, and what remains runtime-enforced?	open	Opened 2026-07-24.
ALF-OQ-006	What compiler invariant and runtime check enforce no nested orchestrator invocation?	open	Opened 2026-07-24.
ALF-OQ-007	Will policy composition be intersection-of-authority, and how will empty policy fail closed?	open	Raised from Lean precedent 2026-07-24.
ALF-OQ-008	Which roots and owners define the finite bootstrap boundary?	open	Relative checking proposed 2026-07-24.
ALF-OQ-009	What evidence is sufficient for mathematical-to-product correspondence?	open	Ledger proposed; adequacy unresolved.
ALF-OQ-010	Which Lean results stay explanatory, generate validators, or become governed evidence?	open	Opened 2026-07-24.
ALF-OQ-011	What is the identity, ownership, reopening, and projection contract for Open Questions?	open	Opened 2026-07-24.
ALF-OQ-012	How can research remain connected to one Plan without folder proliferation?	resolved	Plan registry and one reused formalization node adopted 2026-07-24.
ALF-OQ-013	Are the meta-contract, conformance checker, global invariant set, composition protocol, and bootstrap distinct artifacts or views over one mechanism?	open	Separation introduced 2026-07-24; ownership undecided.
ALF-OQ-014	What proof obligation distinguishes well-formed invariant metadata from actual preservation under transitions?	open	Opened after metakernel critique 2026-07-24.

ALF-OQ-015	Which parts of the Lean kernel/TCB pattern correspond to the product acceptance boundary, and which remain analogy-only?	open	Lean-kernel precedent bounded 2026-07-24.
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Resolved and deferred questions remain visible. This refinement creates a new appendix revision; the next step is to freeze that revision and run staged independent review, not immediate Lean implementation.

system-view Result

- Status: flag
- Target boundary: high-level product and architecture shape for the composable agent-work system
- Stakeholder altitude: project owner and technically literate first-time reader
- Lane handles:
 - surface: Sections 1–2
 - shape: Sections 3–12 and the candidate formalization appendix in Section 16
 - layering: Section 10, expressed as responsibilities rather than fixed layers
 - stances: Section 14
 - alternative_framings: Sections 1–5 and 9–10
 - shape_diagrams: Sections 5–6
 - deferrals: Sections 14–16
- Stances named: thirteen, each routed to a future engineer-view decision row
- Decided-nothing check: pass; candidate shapes and formal constructs are proposals, not product verdicts
- Term-deferral check: flag; the ontology view does not yet exist
- Evidence boundary: conversation-established constraints, local plan/research material, and official Lean documentation for the bounded kernel/TCB precedent; tool options remain an unverified candidate inventory pending research